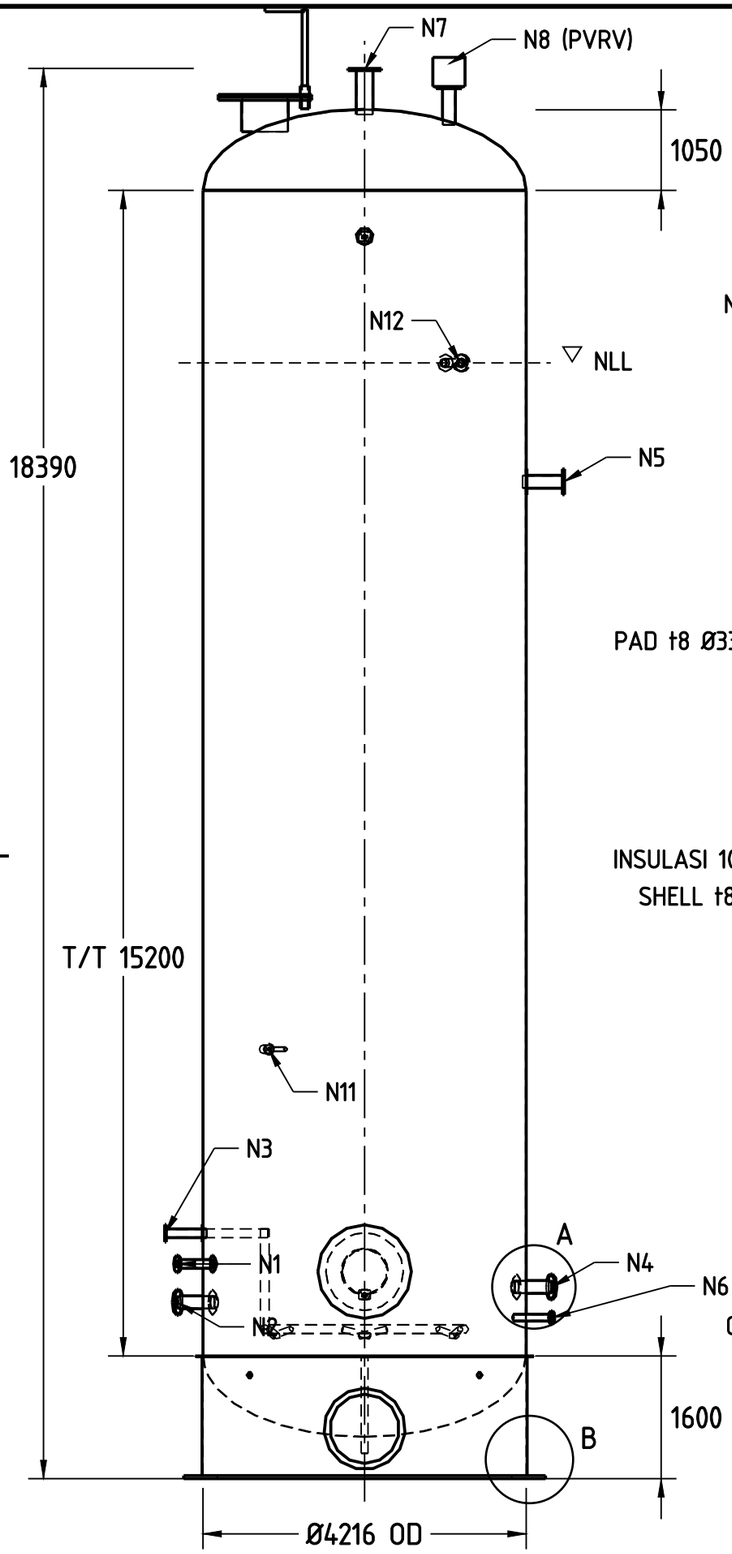
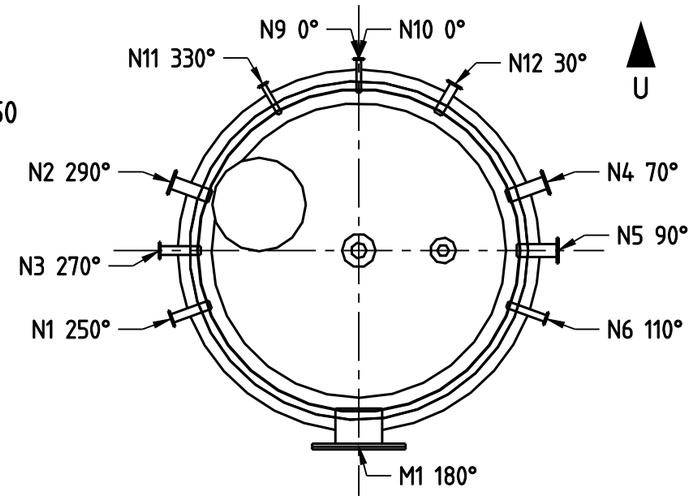
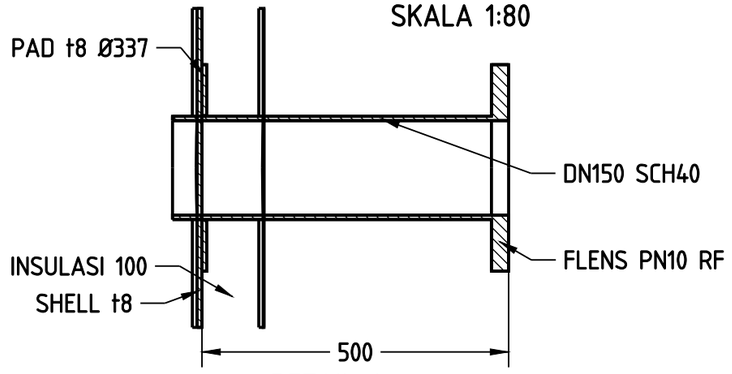


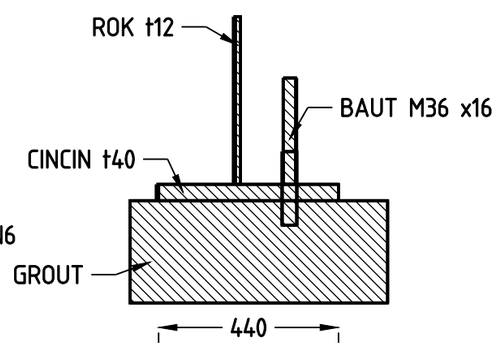
ELEVASI (DARI SELATAN)
SKALA 1:80



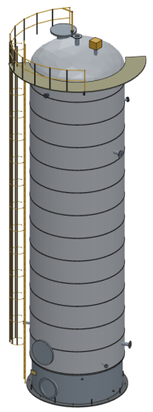
ORIENTASI NOSEL
SKALA 1:80



DETAIL A
SKALA 1:10



DETAIL B
SKALA 1:15



ISOMETRI

JADWAL NOSEL						
Tag	Ukuran	Rating	Layanan	Elev.	Orient.	Pad
N1	DN100	PN10 RF	Umpan reaktor (dari P-201)	2800	250°	Ø229
N2	DN150	PN10 RF	Isap jet mixing (ke P-202)	2300	290°	Ø337
N3	DN100	PN10 RF	Balik jet mixing (header nosel jet)	3200	270°	Ø229
N4	DN150	PN10 RF	Lumpur ke membran (ke P-301)	2500	70°	Ø337
N5	DN150	PN10 RF	Retentat balik (dari MB-301)	13000	90°	Ø337
N6	DN80	PN10 RF	Lumpur buangan (ke P-303)	2100	110°	–
N7	DN200	PN10 RF	Keluaran biogas	puncak	–	–
N8	DN150	PN10 RF	PVRV / relief tekanan-vakum	kepala	–	–
N9	DN50	PN10 RF	Pemancar level LT (atas / bawah)	16200	0°	–
N10	DN50	PN10 RF	Pemancar level LT bawah	2400	0°	–
N11	DN50	PN10 RF	Termometer TT / pH / NH4	5600	330°	–
N12	DN100	PN10 RF	Kaca intip muka cairan	14550	30°	Ø229
M1	DN600	PN10 RF	Lubang orang bawah	2700	180°	Ø1010
M2	DN600	PN10 RF	Lubang orang atas	kepala	–	Ø1010
N13	DN80	PN10 RF	Pengurasan (kepala bawah)	330	–	–

DATA DESAIN	
Data desain	Nilai
Fungsi / jumlah	Reaktor AnMBR termofilik, 14 unit identik + TK-101/TK-301
Kode desain / las	API 650 (acuan) + ASME VIII-1 nosel; las ASME IX
Tekanan / suhu desain	+50 / −5 mbar(g) + hidrostatik penuh; 70 °C (op. 55 °C)
Volume total / kerja	230 m³ / 194 m³, NLL 14000 mm dari BTL
Bahan shell / kepala	SA-516-70N, 8 mm (6 course x 2400), CA 3 mm
Angin / gempa	SNI 1727:2020 (40 m/s) / SNI 1726:2019, Kelas D
Insulasi / kladding	Mineral wool 100 mm 100 kg/m³ / Al 0,8 mm + sabuk
Pelapisan	Dalam: SA 2,5 + epoksi fenolik 400 µm; luar: zinc-epoksi-PU
Massa kosong / operasi	± 20.9 t / ± 219 t
Transport	rebah di low-bed, 17.3 x 4.4 m, 2 trunnion

DAFTAR MATERIAL (MASSA DARI MODEL 3D)				
No	Bagian	Material	Jml	Massa kg
1	Shell (6 course)	SA-516-70N t8	1	12619
2	Kepala 2:1 atas / bawah	SA-516-70N t8	2	2311
3	Rok + rangka akses + vent	SA-516-70N t12	1	2214
4	Cincin dasar + kursi	SA-36 t40	1	1872
5	Baut angkur M36	F1554 Gr.36 galv.	16	23
6	Leher nosel + flens	SA-106B / SA-105	15	1656
7	Pad penguat	SA-516-70N t8	8	50
8	Header jet internal	SS316L DN100	1	936
9	Trunnion angkat	SA-106B DN200	2	155
10	Tangga + sangkar + platform	S275 galv., grating	1	2318
11	Insulasi mineral wool	100 kg/m³	22.1 m³	2212
12	Kladding + sabuk	Al 3003 t0,8	1	675
13	Gasket + baut flens	EPDM / A193 B7-2H	15 set	–

CATATAN :
1. Las butt penuh-tembus; RT spot 10 %, nosel & pad 100 % MT/PT.
2. Uji hidrostatik: isi penuh air 24 jam; uji pneumatik pad 1 bar.
3. Toleransi umum ISO 2768-mK; elevasi dari dasar cincin (EL 0).
4. Orientasi dari utara searah jarum jam; header jet = garis putus.

University: UNIVERSITAS	Checked by : –	Design Lab.	Document type: Equipment Drawing			Rev: 0
	Author : NAMA TIM	Reaktor AnMBR Standar R-201 s/d R-214	Unit: mm	Scale: 1:80/1:10/1:15	Creation date: 15/10/2026	Size: A3
	Eng. Approved by : –		BG-V-201			Sheet number: 2 / 8